# Samuel Vasington

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## EDUCATION

**University of Pennsylvania -** Philadelphia, PA *August 2023 – May 2027*
**M.S.E** Mechanical Engineering: Design and Manufacturing **GPA:** 4.0
**B.S.E** Mechanical Engineering: Mech of Materials, Structure & Design **GPA:** 3.86
**Minors:** Mathematics, Engineering Entrepreneurship

## PROFESSIONAL EXPERIENCE

**Honeywell Fire Mechanical Intern –** Northford, Connecticut *May 2026 – August 2026*

- Conducted static and transient ANSYS failure analysis to identify stress concentrations causing shipping-test fractures; redesigned packaging and internal enclosures and coordinated mold changes with the vendor to eliminate failures
- Developed cabinet, internal enclosures, and chassis for a 2027 NPI product line, creating drawings and ECNs to meet UL requirements; redesigned acoustic features to increase piezo output by 10 dB for Canadian market entry
- Analyzed the Notifier CAB-5 door BOM and created cost analysis using SAP and vendor input; developed three redesign concepts targeting manufacturing cost, labor, and tolerance issues

**Honeywell Fire Mechanical Intern –** Northford, Connecticut *May 2025 – August 2025*

- Designed an injection-molded housing to constrain a communicator module during seismic testing without additional hardware, optimizing for DFM, usability, and production cost
- Developed ESD-safe display board enclosure, ensuring manufacturability and seamless hardware integration; supported by 3D-printed prototyping and UL safety constraints
- Designed Zone Annunciator for Silent Knight and Fire-Lite series panels; redesigned graphic overlay to reduce cost by over 60%, now in production

**Penn Electric Racing (PER) Mechanical Engineer -** Philadelphia, PA *September 2023 – Present*

- Redesigned the rear wing to shift the center of pressure by **7%** and optimized for downforce versus mass; Ran 20+ CFD simulations to optimize the airfoil configuration and endplate design
- Designed an aerodynamic component to improve airflow around the tires and **increase rear wing downforce by 35%**; fabricated a custom mounting method allowing the airfoil to slot and bolt into a fixture mounted on the chassis
- Innovated the club's first device for aerodynamic data acquisition to validate CFD analysis; invented mounting mechanism that was **70% lighter** to collect data in critical positions while **saving 40%** on cost
- Characterized Pitot tube sensors using wind-tunnel airfoil configuration testing across multiple velocities; compared measurements against CFD velocity sweeps; results within **15%** of CFD predictions
- Developed new brake system with redesigned rotors to improve component integration and **reduce mass by 25%**

**Rapid Prototyping Lab and Tangen Studio Technician -** Philadelphia, PA *September 2024 – September 2026*

- Managed daily operations of laser cutters, 3D printers, and additional prototyping equipment
- Provided guidance on student projects using SolidWorks, Rhino, DraftSight and Adobe Illustrator

**Modular Robotics Research Assistant -** Philadelphia, PA *September 2024 – January 2025*

- Engineered and experimented with different propeller prototypes to balance pitch, thrust and moment of inertia, enabling smooth 3-axis movement of a micro-aerial drone

**Academic Project** - Philadelphia, PA *April 2026 – Present*

- Built a life-size, functional BB-8 replica as part of a multidisciplinary team; designed internal enclosures, outer shell components, and mechanical drive systems

## SKILLS & INTERESTS

**Skills:**

- Design and Analysis Tools: *SolidWorks, Creo, ANSYS, Adobe Illustrator, DraftSight*
- Programming: *Python, MATLAB, Java*
- Prototyping & Manufacturing: *3D Printers, Laser Cutters, CNC Machining, GD&T, Milling, Turning*
- Productivity & Collaboration: *SAP, Teamcenter, Jira, Confluence*

**Personal Interests:** *Soccer, Hiking, Formula 1 Racing, Chess*